Suvarcha Narayan
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Risk Management Plan

Techer Biosciences - Stress & Anxiety Reduction Supplement

Approvals

Name	Role	Signature
John Manufacturer	**Principal Manufacturing Lead**	*John Manufacturer*
Joseph Quality	**Principal Quality Control Lead**	*Joseph Quality*
Robert Distributor	**Principal Packaging and Distribution Lead**	*Robert Distributor*

Purpose

This plan will serve as a general protocol for risk management and mitigation during the production process of Techer Biosciences' new supplement, should any of the potential risks identified were to occur during development.

Scope:

Techer Biosciences' new product aims to be a supplement that reduces anxiety and alleviates stress. The design is intended to appear akin to a candy product, for ease of use and transport for the consumer. Recommended dosing has not been confirmed but Techer Biosciences is looking to market this product for those above the age of 3.

References:

Below are a list of relevant protocols and standard operating procedures useful in determination of risk.

- Manufacturing Protocol
- Product Synthesis Protocol
- Protocol for Quality & Standard Check
- Protocol for Batch Documentation, Review, and Record
- Quality Verification Protocol

Roles and Responsibilities

This risk management plan is intended for those involved during the production process of the supplement. This largely includes personnel within manufacturing operations, quality control, and packaging/distribution.

Responsibilities – Manufacturing Operations

- Manufacturing personnel will be responsible for planning and executing risk management & mitigation strategies for potential hazards during the manufacturing process.
- This department will work with quality control to implement strategies that require increased quality checks during manufacturing process

Responsibilities – Quality Control

- Quality control oversight personnel will also be responsible for the design and execution of risk management & mitigation strategies for potential hazards during quality checks.
- Additionally, strategies that may involve increased quality checks throughout the development process will fall on quality control personnel.

Responsibilities – Packaging & Distribution

- Packaging and distribution personnel are responsible for the design and execution of risk management & mitigation strategies for potential hazards during the packaging process of the supplement.
- Any strategies that require increased quality checks during packaging and distribution will have personnel work collaboratively with quality control to ensure risks are appropriately managed and mitigated.

Risk Analysis/Assessments and Requirements for Review

For effective risk management of Techer's production and manufacturing process, a preliminary hazard analysis (PHA) and a failure mode and effects analysis (FMEA) can both provide a comprehensive review of risks throughout the supplement's development process. The PHA will aid in early identification of risks and severities to determine if production of the supplement can continue, while the FMEA will provide greater context during the supplement's production such that relevant actions can be made appropriately (i.e. corrective and preventive action [CAPA]).

To complete these assessments, personnel will need to have complete and detailed documentation of all relevant protocols used in the development process, from synthesis of the product to packaging and distribution steps. Mitigation and management strategies derived from these assessments will require signoffs from involved personnel to ensure the strategies are effective and will be employed. Involved personnel will also be required to hold recorded meetings for these assessments as they are conducted at each phase of production, prior to signoff. The meetings are for administrative purposes and will assist in documentation of overall risk management for the company as a whole.

Criteria for Risk Acceptability

In Techer Biosciences' assessment of risk, personnel should evaluate levels of risk on a qualitative scale, dependent on the severity of the event. To cover all generalities, the following list can be used for evaluation of risk severity:

- Negligible – Inconvenience or temporary discomfort
- Minor – Temporary injury or impairment that does not require medical attention
- Major – Injury or impairment that requires medical attention
- Critical – Significant injury or impairment that is irreversible and permanent
- Fatal/Catastrophic – Causes death

Following the evaluation of severity, additional steps will need to be taken to determine the overall residual risks within the development process of the product. One way this can be completed is to re-evaluate the risk acceptability matrix. If impact is made from implementation of controls, the respective departments can either implement additional controls as means for improvement, or redetermine the benefit-risk analysis.

Activities for Verification of implementation and effectiveness of risk controls.

To evaluate the overall effectiveness of the determined risk controls, Techer Biosciences will need to verify implementation of those controls and document evidence of effectiveness. To confirm that the verification control is implemented, documentation of the quality checks across the development process as well as the reviews of each protocol must be completed. In ensuring that this verification control is effective, testing from quality control, as well as use, design, and process verification protocols must be documented for each product batch/lot.

Activities for production and post-production review

Production and post-production review will periodically be conducted by select personnel of quality control to evaluate the general effectiveness of this risk management plan across all departments. The purpose of these reviews is to document and evaluate whether or not the risk management plan needs to be modified or rewritten. However, this is entirely dependent on whether potential identified risks and hazards change over time across the development process. Refer to quality control and verification protocols.